

RIDDHI GOSWAMI

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[Portfolio Link](#)



PROFILE SUMMARY

Computer Science Engineering student with a CGPA of 9.974 and recipient of the UAE Golden Visa for 'Distinguished Student.' Passionate about Data Science, IoT, and ML/AI applications, with leadership experience in Google Developer Student Club and IEEE. Adept at problem-solving, database management, and leading tech workshops. Skilled in teamwork, leadership, and simplifying complex concepts. Actively seeking internship opportunities to contribute my technical expertise and collaborative spirit.

EDUCATION

B.E. Computer Science Engineering & Minor in Data Science

BITS Pilani, Dubai Campus

CGPA: 9.974

Relevant coursework: Machine Learning, Foundations of Data Science, AI, Data Mining, Object Oriented Programming

Academic Accolades: First in my batch (2023-2024)

TECHNICAL SKILLS

Programming	Python, Java, HTML, CSS, Javascript, C, C#
Databases	MySQL, SnowflakeSQL
Data Analysis	Pandas, Polars, MS Excel
Data Visualisation	Plotly, PyGWalker, Matplotlib, Seaborn
Frameworks	.NET, React, Django, Flask (fundamentals)
Dashboards	PowerBi, Dash, Streamlit, PyShiny
AI & ML	Numpy, Sklearn, OpenCV, Spacy, NLTK, Tensorflow, Keras, LLMs
Electronics & IoT	Arduino, ROS, ROS2

WORK EXPERIENCE

Revenue Optimization and Distribution Intern

Emirates Group HQ, Dubai

JUN, 2023 – AUG, 2023

- Developed apps using **Plotly**, **Dash**, and **Streamlit** for data visualization and analysis.
- Used **Snowflake cloud database** to store, query, and handle large-scale datasets efficiently.
- Individual **ML** Project: Handled a **500M** row dataset, optimizing and parallelizing code. Achieved prediction of flight sell-out dates with a mean absolute error of less than two days.

Software Developer – Structure Team

Jacobs, Dubai

JUN, 2024 – AUG, 2024

- Developed apps using **.NET 8.0 WPF**, **PyShiny** for handling calculations, and visualisations and kept the user interface dynamic and responsive to inputs. Understood how custom components can be built by integrating **Grasshopper API**
- Explored **NLP** models like **Seq2Seq**, **BERT**, and **fuzzy string matching** to enforce corporate naming conventions
- Gained insight into **Azure DevOps**, **Agile** management framework, using **design patterns** and **refactoring**

Software Developer (Upcoming Internship – PS II)

Esri, Sharjah

27TH JAN, 2025 – 4TH JUL, 2025

MISCELLANEOUS EXPERIENCE

Professional Assistant – Computer Architecture Computer Science and Engineering Department, BPDC	SEPT 2024 – current
Management Lead Google Developers Group – BPDC Chapter	SEPT 2024 – current
Event Manager IEEE – BPDC Chapter	SEPT 2023 – AUG 2024
Software Team Member IFOR (Intelligent Flying Object Reconnaissance) – BPDC	SEPT 2023 – AUG 2024

CERTIFICATIONS

More certifications on my [LinkedIn Profile](#)

- HarvardX CS50's Introduction to Artificial Intelligence with Python (Jul 2024)
- Neural Networks and Deep Learning, issued by DeepLearning.AI Academy (Jan 2024)
- Structuring and Managing Machine Learning Projects, issued by DeepLearning.AI Academy (Jan 2024)
- Microsoft Learn AI Skills Challenge (Aug 2023)
- HarvardX CS50x Professional Certification (Jan 2023)
- Postman API Student Expert + Campus Postman Student Leader

PUBLICATIONS AND PRESENTATIONS

- Presented and selected for publication: "Voice-Controlled Bot Navigation: A Novel NLP Approach in Human-Computer Interaction" at the IEEE International Conference on Computational Intelligence and Network Systems (CINS), 2024.
 - Presented my final year project: "AlphaDock: DNA Protein Complex Predictor" before professors for evaluation.
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PROJECTS

More projects on my [Github](#)

Voice Controlled Mobile Bot (Cross Department Research Funded)

This project integrated (Natural Language Processing) **NLP** and the Robot Operating System (**ROS**) to facilitate user interaction with a Turtlebot3, ROSbot XL through voice commands. It employs the **WhisperAPI** and **GPT-4o** to process and understand natural language inputs. It uses **Elevenlabs** for dynamic text-to-speech conversion, demonstrating how the robot can respond to commands vocally based on the input received.

AlphaDock: DNA Protein Complex Predictor

This project involved the identification and analysis of anchor segments within transcription factor binding sites using data from both in vitro and in vivo experiments. Specifically, I used datasets from UniProbe and CIS-BP databases, which include in vitro data from protein-binding microarrays and SELEX. Through this, I have gained experience in **computational biology**, **data wrangling**, and **algorithm** development.

Entity-Relation Identification from Documents

This project taught me how to work with **Spacy** and the Natural Language Toolkit (**NLTK**) to extract entity relations from text paragraphs. The primary objective is to construct detailed **knowledge graphs** that capture the relationships between various identified entities. These knowledge graphs serve as a foundation for translating complex entity relationships, including many-to-one and one-to-many associations, into formal specifications in the **Isabelle-HOL** code.

CSP-based Timetable Generator

This problem of timetable generation belongs to a specific class called Constraint Satisfaction Problem (**CSP**). I developed a custom **Python** tool that processes the class details provided in a PDF file from college and generates possible timetables using **AI** techniques. The tool allows users to input various constraints, such as specific lecture slots or discipline electives, ensuring customised and feasible scheduling.

Automated Detection of Structural Integrity Risks in Buildings

For this project, I first worked on **data preparation**, as there aren't very comprehensive and large sources of images of various defects for model building. I will be working on getting a **local copyright** for this custom dataset of mine. After **cleaning** and **processing** this data, I worked on training various models like **ResNet50**, **ResNet32**, **Inception V3** and state-of-the-art models like **RevCol**, and the transformer based **CoCa** model. I will be working on a journal paper for publishing my findings in this project.

Classifying Catastrophic Events by Type and Severity

I developed **CNN-based** models to classify disaster types (earthquake, flood, hurricane, fire etc) and severity levels using images from social media and UAVs/UGVs. I worked with **ResNet32**, **Inception V3**, and **VGG-16** architectures to achieve high accuracy while maintaining efficiency. Finally, I created a Python-based GUI allowing to users to classify uploaded images and try out the various models.
